question 5 of 5 :

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a) regression equation of y on x

$y = \hat{\beta}_0 + \hat{\beta}_1 x$

where $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

$\hat{\beta}_1 = \dfrac{cov(x,y)}{Var(x)} = \dfrac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n(\bar{x})^2}$

$= \dfrac{152.59 - 11 \times \left(\frac{16.5}{11}\right) \times \left(\frac{100.4}{11}\right)}{28.85 - 11 \times \left(\frac{16.5}{11}\right)^2}$

$= \dfrac{152.59 - \frac{(16.5)(100.4)}{11}}{2\cancel{5}.85 - \frac{(16.5)^2}{11}}$

$= \dfrac{152.59 - 150.6}{2\cancel{5}.85 - 24.75}$

$= \dfrac{1.99}{1.1}$

$= 1.809$

$\hat{\beta}_0 = \left(\frac{100.4}{11}\right) - 1.809 \times \left(\frac{16.5}{11}\right)$

$= 6.413$

$\therefore$ reg line of y on x

$y = 6.413 + 1.809 x$

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reg line of x on y

$$x = \hat{c} + \hat{d}y$$

where $\hat{c} = \bar{x} - \hat{d}\bar{y}$

$$\hat{d} = \frac{Cov(x,y)}{Var(y)}$$

$$= \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum y_i^2 - n\bar{y}^2}$$

$$= \frac{152.59 - 11 \times \left(\frac{16.5}{11}\right) \cdot \frac{100.4}{11}}{923.58 - 11 \times \left(\frac{100.4}{11}\right)^2}$$

$$= \frac{152.59 - \frac{16.5 \times 100.4}{11}}{923.58 - \frac{(100.4)^2}{11}}$$

$$= \frac{1.99}{7.2018}$$

(A) $= 0.2763$

$$\hat{c} = \frac{16.5}{11} - 0.2763 \times \frac{100.4}{11}$$

$$= -1.019127$$

reg line of x on y

$$x = -1.019 + 0.2763y$$

if $x = 1.75$ predicted $y = 6.413 + 1.809 \times 1.75$

$$= 9.578$$

(r)

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b) $r_{yx_1} = 0.755 \rightarrow corr(y, x_1)$ (1)

$r_{yx_2} = -0.811 \rightarrow corr(y, x_2)$ (1)

$r_{x_1x_2} = -0.293 \rightarrow corr(x_1, x_2)$ (1)

i) partial corr coef between y and x_1

$$r_{yx_1 \cdot x_2} = \frac{r_{yx_1} - r_{yx_2} r_{x_1x_2}}{\sqrt{1 - r_{yx_2}^2}\sqrt{1 - r_{x_1x_2}^2}}$$

$= 0.926$ (2)

partial corr coef between y and x_2

$$r_{yx_2 \cdot x_1} = \frac{r_{yx_2} - r_{yx_1} r_{x_1x_2}}{\sqrt{1 - r_{yx_1}^2}\sqrt{1 - r_{x_1x_2}^2}}$$ (2)

$= -0.941$

ii) Multiple corr coef between y and x_1, x_2

$$R_{y \cdot x_1x_2} = \sqrt{\frac{r_{yx_1}^2 + r_{yx_2}^2 - 2 r_{yx_1} r_{yx_2} r_{x_1x_2}}{1 - r_{x_1x_2}^2}}$$

$= 0.975$ (3)

$R^2_{y \cdot x_1x_2} = 0.9512$